

SCIENCE

CHAPTER 1

Science as a Way of Knowing

Lesson 1

The Scientific Method

Do List

1. Explain the importance of the scientific method.
2. Enumerate the steps of the scientific method.
3. Solve a common problem using the scientific method.

Energize Your Mind

Have you been sick recently? Did you visit a doctor? Doctors cannot determine right away why you are sick. They have to observe you first and look for symptoms that may indicate the presence of a disease. Various tests are then conducted to help them assess your health condition. After the disease is identified, the appropriate treatment is given. By following an organized process or method, doctors are able to treat health problems and save lives.

Do you also use a method in solving problems? Do you observe and perform tests before you decide on something? Whether for medical issues or simple household or school concerns, it is helpful to solve problems in a logical way. This logical problem-solving approach is known as the scientific method.

What is the scientific method? Have you ever used it to solve everyday problems? Do the following activity and see if your approach to problem solving follows the scientific method.

Explore Around

PAPER ENGINEERING I

What You Need

old newspaper
3 books

What to Do

1. Using only the newspaper, make a vertical structure that is strong enough to carry the weight of your books. Decide on the shape or form and thickness of your structure. It should be at least

2 inches tall.

2. Take note of all of your actions, even the mistakes that you committed, while making your structure.

What You Have Learned

Answer the following questions. Write your answers in the space provided.

1. What steps did you take in making your structure?
2. What is the shape of your structure? Why did you choose such a shape?
3. How strong do you think is your structure? Why do you say so?

How You Learned

1. Was the activity challenging? Why?
2. What difficulties did you encounter while doing the activity?

Equip Yourself

A scientific investigation is usually triggered by an observation, that is, some-thing that catches a scientist's attention. For instance, an environmental scientist might notice that the temperature of the Earth has been rising for the past consecutive years and he/she would like to know why. An entomologist (one who studies insects) observes that the number of bees in a certain field is declining and sets out to find out the reason behind it. How do they follow up or look deeper into their observations? How about you? How would you follow up on your observations of the physical world, especially of a problem in your environment?

The scientific method is a logical way of answering questions or solving problems based on observable or measurable evidence. It is a set of procedures that helps scientists gain new knowledge and update their knowledge of the physical world. It involves systematic observation, measurement, communication, formulation of hypothesis, and experiments. The application of the scientific method is essential in doing science so that scientists can track the progress of their investigation and not miss any important detail or procedure. Applying the scientific method can also help reduce the errors that scientists may make when conducting investigations. Following the scientific method also saves them time because repetition of steps and unnecessary actions can be avoided.

1. Ask a question.
2. Do research.
3. Conduct an experiment.
4. Form a hypothesis.
5. Gather and analyze data.
6. Draw a conclusion.

the scientific method

Even if you are not a scientist, you can also make use of the scientific method encountered every day. The scientific method consists of a number of key steps. In fact, you may have already used it to solve simple and complex problems that you

Asking a Question

The scientific method begins with a question or a problem. It can be anything that you had observed or experienced and want to know more about. For instance, you overheard someone saying that placing black pepper on areas infested by rats will make these animals go away. Wondering whether the statement is true, you ask yourself the question: "Do rats dislike black pepper?"

Doing Research

After asking a question or stating a problem that you want to solve, you continue the process by doing research or by gathering existing information about your topic. You may get information from books or on the internet, or you may ask an expert. Make sure to gather information only from reliable and credible sources. When searching the internet, look for academic journals or resources. Compare the opinions of scholars or experts about your topic so that you can evaluate and verify the information you have gathered. Doing research is very important because it gives you insights into the topic. It also prevents you from repeating the errors of people who might have asked the same question or have tried solving the same problem in the past.

Forming a Hypothesis

After doing research, you can now form a hypothesis. A hypothesis is a potential answer to a question or problem at hand. It is a possible explanation for how things work or came to be on the basis of limited, measurable evidence. You can make predictions to prove or disprove your hypothesis. A properly worded prediction is often presented in the form of an if-then statement. One example of a prediction is as follows: "If I place black pepper on the rat-infested areas of our house, then the rats will go away." A good hypothesis is testable, which means it can be proven right or wrong.

Conducting an Experiment

You need to conduct an experiment to test if your hypothesis is correct. A methodology is a system of procedures that describes what you plan to do and Before conducting an experiment, design your experiment or methodology first. identifies the data that you aim to collect. To do this, begin with identifying your independent variable (the factor to be purposely manipulated in the experiment) and your dependent variable (the factor that is predicted to change based on how the

independent variable is manipulated). In our example, the independent variable is the presence or absence of black pepper in areas infested by rats. The dependent variable is the presence of rats in the infested areas. The methodology also provides a systematic description of how you will change the independent variable and observe its effects on the dependent variable.

The materials that you will use in the experiment should also be identified. Once the methodology and materials are ready, you can begin your experiment. For our example, you can begin the experiment by placing a generous amount of black pepper in two high-activity, rat-infested areas in your house and observe what will happen. Do several trials to make sure that you get enough data and to validate the accuracy and precision of the obtained results. This will make the result of your experiment conclusive and reliable.

Gathering and Analyzing Data

During the experiment, record all your observations. These observations are your data. Data are pieces of information that you collect from the experiment and which will help prove or disprove your hypothesis. You must be precise and accurate in measuring and recording information from your experiment to generate reliable, valid results. Once you have gathered data, analyze them by carefully looking for any patterns, connections, and inconsistencies.

In the table below, it can be observed that there were rats in all specified areas. Specifically, the most number of rats were found in the storage room.

Trial	Kitchen Cabinet		Storage Room	
	With Black Pepper	Without Black Pepper	With Black Pepper	Without Black Pepper
1	1	2	3	1
2	1	2	1	1
3	2	1	2	3

Drawing a Conclusion

After analyzing the collected data, compare the results of the experiment to your hypothesis and then draw a conclusion. A conclusion is a judgement based on facts or on the results of an experiment. Your conclusion will confirm if your hypothesis is correct or not. In the example, the conclusion would be, "Rats did not go away after black pepper was placed in the infested areas."

If the conclusion does not support your hypothesis, you may form another hypothesis and make changes in the design of your experiment based on what you have learned during the experiment. This then brings you to repeat some steps of experiment to ask new questions and form new hypotheses, is called the iterative the scientific method. This feedback process, which involves using the results of an *process*.

The scientific method can be used in all fields of science. In fact, the gadgets that you enjoy today are the results of studies and efforts done in the field of science and technology that used the scientific method. However, not all scientists follow are areas of science that require a different system of investigation. For instance, scientists who study the evolution of the first reptiles on Earth cannot travel back in time by millions of years to make direct observations and

test their hypothesis. In such cases, scientists modify their use of the scientific method. However, the goal remains the same: to ask a question, gather and analyze data, and draw conclusions.

The scientific method is a logical way of solving problems based on observable or measurable evidence.

The key steps of the scientific method are asking a question, doing research, forming a hypothesis, conducting an experiment, gathering and analyzing data, and drawing a conclusion.

An independent variable is the factor to be purposely manipulated in the experiment whereas a dependent variable is the factor that is predicted to change based on how the independent variable is manipulated.

After learning the steps of the scientific method, you can now work on problems but be sure to follow the steps of the scientific method this time. in a more systematic and logical way. Do the Paper Engineering activity again

Explore Around

PAPER ENGINEERING II

What You Need

3 books
old newspaper

What to Do

The challenge is the same as the first activity on page 6, but this time, you have to follow the steps of the scientific method. Think of how you can incorporate what you have learned in building a much effective structure.

What You Have Learned

Write your answers in the space provided.

Describe what you did in the following steps of the scientific method.

1. Asking a question:
2. Doing research:
3. Forming a hypothesis:
4. Conducting an experiment:
5. Gathering and analyzing data:
6. Drawing a conclusion:

How You Learned

1. How does your new structure differ from the previous structure that you made? Do you think it is better? In what way/s?

2. How did the scientific method help you in making the new structure?

Express Your Learning

Answer the following questions:

1. What are the steps of the scientific method?
2. What is the importance of the scientific method?
3. How can you use the scientific method in your daily life? Cite one example.
4. How did you feel about the lesson? Explain your answer.

Lesson 2

Scientific Tools

Do List

1. Identify and describe some scientific tools.
2. Demonstrate proper use of some scientific tools.

Energize Your Mind



Have you ever visited a laboratory? Have you ever used any scientific tools that are similar to what the girl in the photo above is using? Scientific tools are used for different purposes, such as observing and measuring objects. These tools are often used in experiments in laboratories. Some of these tools can also be used in household and business tasks. For instance, your parents monitor your height using a measuring tape and determine your weight using a weighing scale. When you have a fever, they use a thermometer to measure your temperature. Supermarket vendors also use a weighing scale to measure the mass of their goods. Thus, scientific tools are helpful not only in learning science but also in performing daily activities. Do the following activity to learn more about scientific tools.

Explore Around

What You Need

TOOL HUNT

notebook

pencil

What to Do

1. Under each category in the table below, draw two (2) tools that you can find in your home.

Place

Observing Tools

Measuring Tools

Heating Tools

home

2. Visit the science laboratory in your school. Draw two (2) laboratory tools under each category.

Place

Observing Tools

Measuring Tools

Heating Tools

science laboratory

that you have seen there.

3. Recall a trip to the supermarket. Under each category, draw two (2) tools

Place

Observing Tools

Measuring Tools

Heating Tools

Supermarket

What You Have Learned

Answer the following questions. Write your answers in the space provided.

1. Which tools are commonly found in your home and in the supermarket?
2. What are the tools that you have seen in the science laboratory?
3. What measuring tools are present in all three places? How are they similar to one another?

How You Learned

1. How did you feel while doing the activity?
2. Did you have difficulties classifying the tools according to their category? Explain your answer.

Equip Yourself

After learning about the scientific method, you are now aware of the importance of gathering and recording accurate, precise information. In this lesson, you will learn about some common

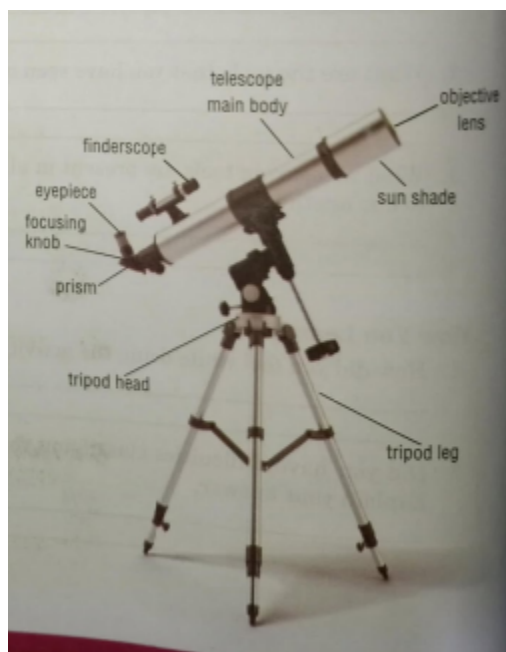
scientific tools that are used by scientists to investigate the natural world. Scientific tools are instruments used by scientists to make observations, take measurements, and conduct experiments. These instruments are grouped according to their use. Knowledge about these tools will help you conduct your own experiments in the near future.

Observing Tools

Can you accurately describe the body structure of an ant using only your eyes? If you just use your eyes to observe an ant, you can say that it is small and black or red. However, it would be impossible for you to see and describe its sensory organs because those details are too small for your eyes to see. Human senses have limitations and cannot be completely relied on to provide accurate and precise information. Thus, you need to use specialized tools to extend the capability of your senses and improve your observations of objects that are extremely small or distant. Some of the common observing tools are the telescope, magnifying glass, and microscope.

Telescope

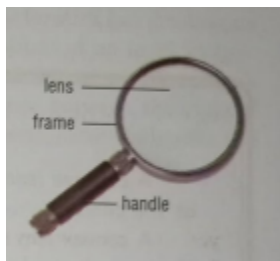
A telescope is an instrument used for viewing distant objects such as the moon and the stars. The telescope is said to have been invented by Hans Lippershey (1570-1619), who applied for a patent for the device in 1608. According to Lippershey, the device, which has a contact eyepiece aligned with a convex objective lens, can magnify distant objects three times. The telescope works by magnifying the image of distant objects through the use of mirrors and lenses. There are two types of telescopes. These are the reflecting telescope and the refracting telescope. The reflecting telescope uses a parabolic mirror to collect and focus the parallel light



rays coming from distant objects such as celestial objects. The refracting telescope, on the other hand, uses lenses to gather more light than what can be absorbed by the human eye. These lenses are the objective lens and the eyepiece lens, which make objects from afar look brighter, clearer, and magnified.

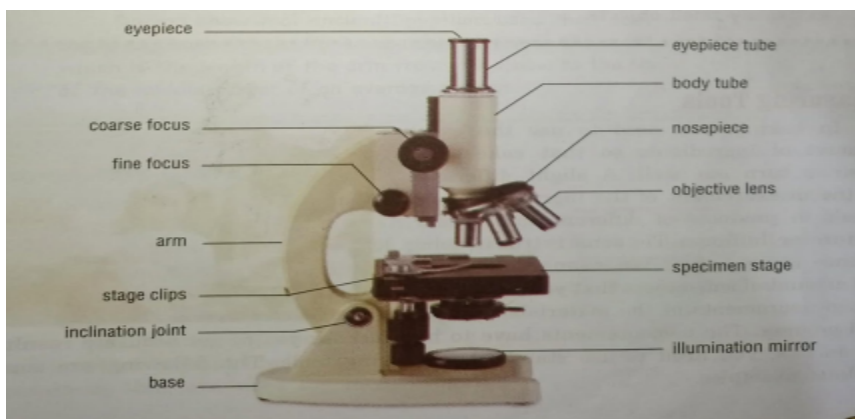
Magnifying Glass

The magnifying glass or the handheld lens was invented by Sir Roger Bacon (1214-1294), a lecturer at Oxford University in England. It is a device that is used to view small or minuscule objects. Most magnifying glasses use double convex lenses that allow light rays to bend toward the center of the lens. When the magnifying glass is placed close to an object, the object looks larger than what it appears to be to the human eye. The magnification of a magnifying glass is determined by its distance from the object being viewed.



Microscope

A microscope, which is believed to have been invented by Hans Janssen and Zacharias Janssen (1585-1632), is a device that allows you to see enlarged images of very small objects. Like the magnifying glass, the microscope makes use of lenses. As you look through the eyepiece, the light from the object that you are looking at is bent or refracted toward your eye, making the object look bigger. Convex



compound microscope. A compound microscope uses two convex lenses with short focal lengths. There are different types of microscopes. The most common of these is the compound microscope. When you look into the eyepiece, you will see that the object is magnified and inverted at the same time. A compound microscope has a magnification of 1000x.

Extend Your Understanding

(Word Unlock)

A concave lens is a lens that is thinner at the center.

A convex lens is a lens that is thicker at the center.

concave lens

convex lens

Tools for observing, such as telescopes, magnifying glass, and microscopes, help provide accurate and detailed observations of extremely distant or extremely small objects.

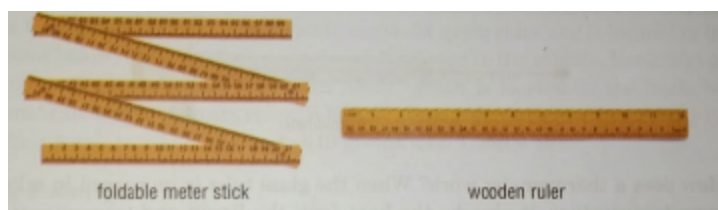
Measuring Tools

In baking, you need to use the exact amount of ingredients so that cakes and pastries turn out well. A slight difference in the measurement of the ingredients may result in products of different taste, color, texture, or fluffiness. The same is true in doing science experiments. You cannot just guess the amount of substances that you will mix or the measurements of the materials that you will prepare. The measurements have to be exact for you to get accurate results. To do this, you need to use standard measuring tools. The following are some common examples.



Meterstick and Ruler

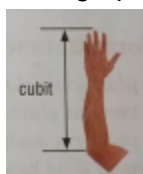
A meterstick and a ruler are tools for measuring the lengths or widths of objects or the distance of one object to another. These measuring devices may be made of wood, metal, or plastic. A meterstick is used to measure the length or width of objects, such as your classroom or the school's court and long distances such as that between the ends of a basketball court. One side of the meterstick has gradations of up to 100 centimeters, whereas the other side has gradations of up to 39 inches. Meanwhile, a ruler is used to measure the length or width of objects, such as a book or a chopping board, and short distances such as the distance between the ends of an eraser. One side of the ruler has gradations of up to 30 centimeters, whereas the other side has gradations of up to 12 inches.



Extend Your Understanding

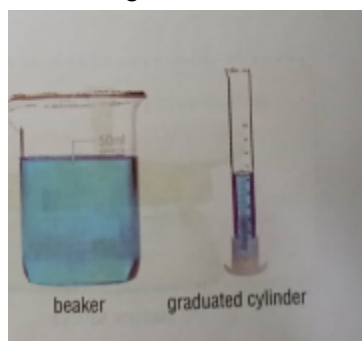
(Know Zone)

Do you know that ancient Egyptians used their body parts as measuring tools? An example of this is the cubit, which is the length of the arm from the elbow to the tip of the middle finger of an average person.



Laboratory Glassware

Laboratory glassware are objects that are usually made of glass and are used in scientific laboratories. Examples of glassware used for measuring substances are beakers, which are also used for mixing and transporting fluids, heating fluids over an open flame, and containing chemicals during a reaction. Graduated cylinders are tall, narrow containers that are used for measuring the



volume of liquids. Most glassware used in laboratories are made of borosilicate glass, which type of glass can also contain acidic or corrosive chemicals. However, glassware are fragile and break easily and thus should be handled carefully.

Thermometer

A thermometer is an instrument that measures temperature. Temperature is the measure of hotness or coldness of a substance or object. There are several types of thermometers. One is the laboratory thermometer, which is commonly used for science experiments. Laboratory thermometers may be partially filled by either mercury or alcohol. However, for the past several years, mercury has no longer been in use in thermometers because the substance has been found to be toxic.

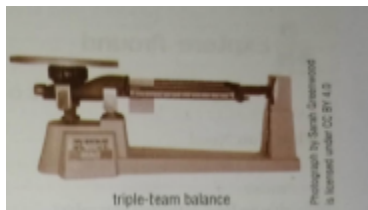
How does a thermometer work? When the glass tube is immersed in a liquid of unknown temperature, it absorbs the heat from the liquid and transmits it to the substance inside the thermometer. The substance then rises or drops depending on the temperature of the liquid.

Platform Balance

A platform balance is a tool for measuring mass. It is an equal-arm balance in which two flat platforms are attached to the top side of the beam, one at each end. A platform scale is used to weigh people or very heavy objects. It consists of a platform or solid base as well as a series of levers and balances that are used together to compute the total mass of the object or person being weighed.

Triple-Beam Balance

The triple-beam balance is another tool that is used to measure mass. This measuring tool is composed of two parts: the base and the pan. The base is a long metal platform that supports the measuring tool. The pan is the metal platform that rests on the top of the base and where the object to be weighed is placed. You determine the mass of the object being weighed with the use of the three beams, each with a rider that you move along the beam. The first beam, which is the lightest, has a 10-gram scale and a 0.1-gram rider and is located in front. The second beam, which is the heaviest and is located in the middle, has a 500-gram scale and a 100-gram rider. The third beam, which is located in the back, has a 100-gram scale and a 10-gram rider. When all three riders are positioned fully to the right, their weights would add up to 610 grams ($500 + 100 + 10$).



Spring Balance

A spring balance is a tool that is used to measure the weight of an object. It consists of a spring and a hook from which the object to be weighed is hung. The unit of measurement for weights that are measured using a spring balance is newton (N). A load applied on the spring causes the spring to stretch. This, in turn, causes the crossbar to move and rotate the pinion and the load-indicating pointer.

There are two types of spring balances: the crane scale, which is suspended from a crane hook, and the fish scale, which is used in homes.



The following are some common scientific tools that are used for measuring:

meterstick and ruler for length

beaker and graduated cylinder for volume

thermometer for temperature

platform balance and triple-beam balance for mass and spring balance for weight

Explore Around

SAME OR NOT?

What You Need

small jar

water

dye or food coloring (any color)

100 mL beaker

100 mL graduated cylinder

What to Do

1. Fill the small jar with water. Add a small amount of dye or food coloring to the water and mix them together using the spoon.

2. Pour half of the colored water into the beaker. The level of the liquid should fall on a line or mark of the beaker. Record the volume.

40

a. Pour the contents of the beaker into the graduated cylinder. At eye level, determine under what level the liquid falls and observe the curve at its surface. The reading will be based on the meniscus, the lowest

part of the curve 4. You may repeat the procedure to practice measuring different volumes.

What You Have Learned

Answer the following questions. Write your answers in the space provided.

1. What is the volume of the liquid in the beaker?

2. What is the volume of the same liquid in the graduated cylinder?

3. How would you compare the two measurements?

4. Why do you think the two measurements differ from each other?

How You Learned

1. How did you feel about the activity?

the/se problem/s? 2. What problem's did you encounter during the activity? How did you handle

Heating Tools

At home, you may be using a gas stove or a microwave oven to heat or cook your food. In the laboratory, you also need sources of heat to conduct scientific studies. Examples of heating tools that are used in laboratories are alcohol lamp, Bunsen burner, and the hot plate



temperature. The An alcohol lamp uses ethyl alcohol to produce a flame of low length of the flame can be controlled by adjusting the length of the wick The Bunsen burner, which was named after its inventor, Robert Bunsen, uses either natural gas or liquefied petroleum gas to produce a flame that can be adjusted. It consists of a metal tube that is attached to a base with a gas inlet at the lower end of the tube. The lower end may have a valve that can be turned to adjust the amount of gas that goes into the tube.

Unlike the Bunsen burner and the alcohol lamp, a Aat plate ha alat surface surface on which the object to be heated is placed. The temperature of the flat the melting or boiling point of a substance. Because the lot plate uses electricity. it is less harmful than a Junsen burner or an alrshol lamp.

Express Your Leaming

Answer the following questions:

1. What are the different scientific tools?
2. Your teacher asked you to boil 45 mL of water. What materials will you use and why?
3. Is knowing about tools and their uses important in your day-to-day activities? Why?

How did you feel about the lesson? Explain your answer:

Lesson 3

Being a Scientist

Do List

1. Describe and practice different scientific attitudes.
2. Identify some notable scientists and their significant contributions.

Energize Your Mind



Being a scientist is not just about discovering answers to countless questions. The work of scientists does not stop after a discovery has been made. The attitudes that they exhibit while planning and working on an experiment are as important as those that they show when presenting their findings for public questioning. Moreover, as you learn about some famous local and foreign scientists, you will find that these attitudes have proven essential in achieving success in any chosen field. Work on the next activity and discover what attitudes you have developed through the years as well as the traits that you can improve on.

Explore Around

SCIENTIFIC ATTITUDES CHECKLIST

What You Need

pencil

What to Do

Accomplish the checklist below. Put a check mark in the box if your answer to the question is yes. Put a cross mark if your answer is no.

Attitude

Description

keenly observant

Do you spend a lot of time observing things around you?

Are you sensitive to changes in your surroundings?

Are your estimates sharp?

patient

Can you extend your time in waiting for her or someone?

Are you not bored easily when doing a task?

Can you endure difficult circumstances?

curious

Do you like to ask questions?

Do you always want to know about the meaning of things or phenomena

Are you eager to find out earliest new things or do e ^ (SO(JM))

open-minded

Do you accept other people's views about things?

observations! Do you take into consideration other people's

Are you ready to modify your beliefs and opinions when new facts and information show more accurate and updated results?

rational

Are your decisions primarily based on reason?

Do you base your opinions on facts instead of on hearsays and superstitions?

Do you search for evidence before establishing your claims?

critical analytical

Do you question and challenge concepts or ideas?

Do you consider first all options and possibilities before deciding on anything?

creative

Do you listen and carefully evaluate all arguments of an issue before making a judgment?

Are you always full of ideas?

Are you good at imagining things?

Do you usually think of more than one way of presenting something?

resourceful

Do you find use for scrap materials?

Do you recycle materials?

Can you produce an output despite the lack of specified materials?

truthful

Are you sincere while responding to people's questions even if what you say may displease them?

Do you honestly report the results of your experiments even if they contradict your hypotheses?

Do you reject any form of cheating?

humble

If you have errors in your work, do you acknowledge them?

Can you take corrections from other people?

Can you accept that other people may provide better ideas than you?

What You Have Lurned

Answer the following questions. Write your answers in the space provided 1. Which scientific attitudes do you power?

2. How are you able to demonstrate your scientific attitude?

3. Why do you think these traits are important?

How You Learned

1. Which scientific attitude is your strongest trait? Why do you say so

2. Which scientific attitudes do you need most of the time? Why?

3. Which scientific altitudes do you want to develop? Why?

Did you check many boxes in the checklist on pages 26-267 If you did, then you most probably have the traits of a scientist. Try to develop and practice these attitudes more. Even if you are not yet a scientist, these traits can help you become efficient in doing your daily tanka

Equip Yourself

Scientific Attitudes

Like many other professions, being a scientist requires a certain level of standard. Scientista deal with highly conscious mental activities. To be able to perform your duties effectively, you should have deep knowledge about a chosen scientific field. Moreover, having a good background in mathematics is always an advantage. However, more than having knowledge, you must also acquire and practice appropriate scientific attitudes as you do science. These attitudes, along with your established skills, will help you become successful in your chosen field.

Scientific attitudes are traits that are demonstrated by scientists in their work-From coming up with a problem statement to drawing and presenting results and a conclusion, scientists show a set of attitudes. These attitudes guide scientists in perfecting their inventions and discoveries and encourage them to explore other ideas and possibilities. Moreover, scientific attitudes concern a person's perception of his or her surroundings his or her take on what is true ard logical, and his or her moral values

Different types of activities develop certain scientific attitudes. The setup of these activities, such as whether these are to be done alone or in groups, also determine what attitudes can be most developed. While you can practice patience, open-mindedness, and humility while working alone, the development of these traits is given more focus when dealing with different personalities and ideas.

Recall your experience in making a structure using old newspapers. Can you identify which attitudes you and your group mates displayed while creating two versions of the structure? These attitudes surely helped your group come up with a successful output. Attitudes, such as being open minded, humble, and patient, strengthen teamwork and make working together more enjoyable.

The previous activity also showed that scientific attitudes are important in every step of the scientific method. From determining the problem statement to analyzing the data that you have gathered, it is important to be observant and to be alert for changes in your subject matter and in the environment. You also have to be critical even of your own findings. You should always double check and verify your calculations and investigate unusual occurrences.

When finalizing, printing or by acknowledging any that you make and by being grateful for other people's input

Scene attitudes are traits that sets display in their work These attitudes range from motor values fruits that deal with a person's perception of his or her surroundings and what is logical and true

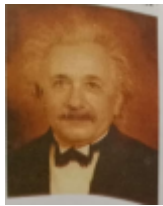
Notable Scientists

During the early times, philosophy came before science. These were poets, only philosophers, who loved wisdom and tried to explain how and why things happen the way they do. Without the aid of tools and instruments, while training and knowledge to explain the natural world. No view human thinking advanced, and the methods used by $\mu\pi\omicron\rho\lambda\alpha\ \tau\alpha\ \acute{\epsilon}\tau\eta\ \mu\upsilon\sigma\eta\mu\epsilon\nu\alpha$ beramind. Today, people have a much deeper and understanding of the nature, structure, and behavior of different The of events in their natural surroundings

A number of notable scientists have added important contagion to society. Try to identify the terrific attitudes that they have exhibited out their car

Foreign Scientists

The scientific achievement of several scientists texts monumental that they continue to impart today's advancements in neurology and life in general. Some of the most influential scientists are Stein, Stephen Hawking, and Antane van Leeuwendek What is the life of a wenland like? How their contest ended Albert Ein hutrans influenced people's



Albert Einstein (1879-1955) was a German-born theoretical physicist. He published over 300 studies, including those that tackle complex theories in physics. Einstein had an overwhelming curiosity about the mysteries of physics. Secure performing experts at certain complex theories was impossible during his time, being used "thought experiments to solve physics.

Named the Person of the Century in 1999 by Time magazine, Einstein has influenced people in human history. He is renowned with genius and has since been regarded as one of the most



Stephen Hawking (1942-2018) was an English theoretical physicist and astronomer. His studies explained that black holes are not completely black but instead emit radiation. This finding challenged the theory that nothing can escape black holes. His discovery demonstrated that matter, in the form of radiation, can escape the gravitational force of a black hole. At age 21, Hawking was diagnosed with amyotrophic lateral sclerosis, a motor neuron disease that bound him to a wheelchair for the rest of his life. However, this condition did not stop him from doing groundbreaking work in physics and cosmology. He earned numerous prestigious recognitions, including the Albert Einstein Award, Wolf Prize, and Prince of Asturias Award. Hawking's perseverance in his pursuit of knowledge made him one of the greatest scientists of his time.



Antonie van Leeuwenhoek (1632-1723) was a prominent Dutch microbiologist who in Leiden improved the microscope. He was known to have made around 500 microscopes. Using these microscopes, he was able to discover bacteria, sperm cells, and red blood cells. These discoveries and other several microscopic observations won him the title Father of Microbiology. A pastor remarked after Leeuwenhoek's death that the "diligence and tireless labor that the scientist displayed in making excellent lenses led him to discover the many secrets of nature."

Local Scientists

Several Filipino scientists have been recognized for their significant contributions that address the needs of Filipinos. Some famous local scientists include Paule Campos, Dolores Ramirez,



and Lourdes Cruz. **Paulo Campos** (1921-2007), dubbed as the Father of Nuclear Medicine in the Philippines, established the country's first medical research laboratory at the College of Medicine of the University of the Philippines in the 1950s. He also pioneered the construction of a radioisotope laboratory in the country. This paved the way for the study and eventually, the discovery, of an effective way to combat goiter by injecting patients with iodine oil, a treatment.

ana National Scientist in 1989 by the late former President Corazon Aquino. that was hacked by the World Health Organization(WHO). Dr. Campos was recognized



Dolores Ramirez is a biochemical geneticist who studies, diagnoses, and treats metabolic diseases) and a cytogeneticist (one who studies the relationship between chromosomes and cell behavior and identifies abnormalities in chromosomes). She is known for her work on the chromosomal properties of local crops such as rice, coconut, banana, and sugarcane. She also studied ornamental plants and fruits such as durian, lanzones, and santol. One of her most important contributions to science and society was her investigation of the genetics of the macapuno endosperm of coconuts. Her investigation made macapuno production a good source of income for Filipinos in the rural areas. Ramirez's contribution to the development of genetics in the country earned her the title of National Scientist in 1998.



Lourdes Cruz is a scientist with an expertise in the field of biochemistry, which utilizes chemistry concepts in studying biological processes. She has published over a hundred scientific journal articles, including investigations on predatory snails. The papers published on these snails tackled their venoms' properties, particularly the toxins they contain. Her findings, specifically on conoscias (substances present in the venom of cone snails), became very significant in understanding brain activities. In 2001, Dr. Cruz initiated the establishment of the Rural Livelihood Incubator, which provided. Bataan. She was proclaimed a National Scientist in 2006. economic, medical, and educational programs for the Aeta community in Morong. Can you name other scientists? What have they contributed to the field of science? Read more about these scientists and others and learn from their lives. You might be the next Einstein or Dr. Campos of your generation

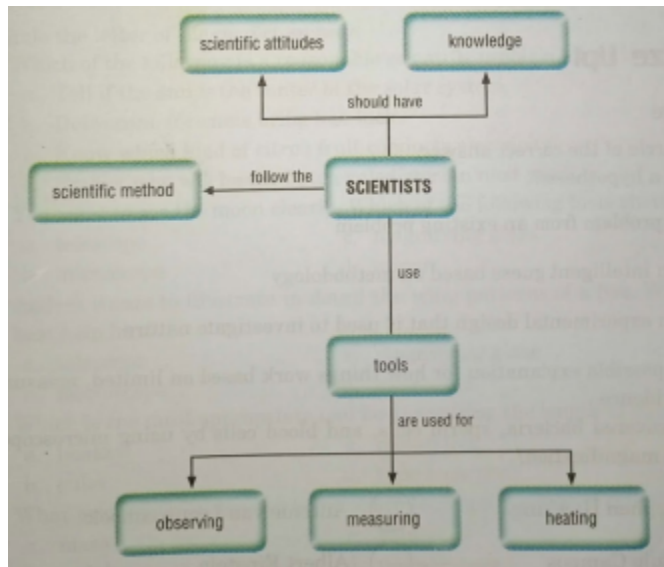
Some of the great foreign scientists include Albert Einstein, Stephen Hawking, and Antonie van Leeuwenhoek. Some of the most exceptional Filipino scientists are Paulo Campos, Dolores Ramirez, and Lourdes Cruz

Express Your Learning

Answer the following questions:

1. Aside from knowledge, what else should a scientist possess? Why?
2. Choose one scientific attitude. How does possessing such a trait help scientists make scientific breakthroughs? Explain your answer.
3. How did you feel about the lesson? Explain your answer.

Map It



Wrap Up

The scientific method is a logical way of solving problems based on observable or measurable evidence. The steps of the scientific method include asking a question, doing research, formulating a hypothesis, conducting an experiment, gathering and analyzing data, and drawing a conclusion. Examples of scientific tools that are used to make accurate observations of small and distant objects are the telescope, magnifying glass, and microscope. Some scientific tools that are used for taking accurate measurements are the meterstick and ruler (length), beaker and graduated cylinder (volume), thermometer (temperature), platform balance and triple-beam balance (mass), and spring balance (weight). Some scientific tools for heating are alcohol lamps, Bunsen burners, and hot plates. Scientific attitudes are traits that scientists display in their work. These traits range from moral values to those that deal with a person's perception of his or her surroundings and of what is logical and true.

Size Up

Knowledge

Shade the circle of the correct answer.

1. What is a hypothesis?

a problem from an existing problem

an intelligent guess based on methodology

an experimental design that is used to investigate nature

a possible explanation for how things work based on limited, measurable evidence

2. Who discovered bacteria, sperm cells, and blood cells by using microscopes of supreme magnification?

Stephen Hawking

Antonie van Leeuwenhoek

Paulo Campos

Albert Einstein

3. In which step of the scientific method do you decide whether to accept or reject your hypothesis?

analyzing the data

drawing the conclusion

collecting the data

conducting the experiment

4. Which tool should you use to measure weight?

Beaker

spring balance

thermometer

platform balance

5. "The liquid is 100°C. What property of matter is being measured in this observation?

mass

volume

weight

temperature

Process/Skill

Encircle the letter of the correct answer.

1. Which of the following is a reasonable scientific problem to work on?

- a. Tell if the sun is the center of the solar system.
- b. Determine if comets bring bad luck.
- c. Know which kind of citrus fruit conducts electricity.
- d. Predict who will be the batch valedictorian next year.

2. You want to see the moon clearly. Which of the following tools should you use?

- a. telescope
- c. magnifying glass
- b. microscope
- d. Mirror

3. Andres wants to illustrate in detail the wing patterns of a bee. Which tool will best help him?

- a. telescope
- e. magnifying glass
- b. microscope
- d. Eyeglasses

4. Which is the most appropriate tool for measuring the length of your pencil?

- a. beaker
- c. meterstick
- b. ruler
- d. tape measure

5. What property of water can be measured using a graduated cylinder?

- a. mass
- c. volume
- d. temperature
- b. Weight

6. Your teacher asked you to measure 850 mL of water for your experiment. Which measuring tool should you use?

- a. medicine dropper
- c. ruler
- b. magnifying glass
- d. Beaker

7. Shaina wants to heat a glass tube so she can bend it. Which tool should she use?

- a. hot plate
- c. platform balance
- b. Bunsen burner
- d. gas stove

8. Instead of using lemon, Joanne used calamansi as an alternative in the experiment she is working on. What scientific attitude does she exhibit?

- a. patience
- c. resourcefulness
- b. creativity
- d. Rationality

9. Refusing to believe in the superstitions told by folks in Barrio Salvador, Dr. Chua set up a team to study the outbreak of an epidemic in the town. What scientific attitude does he exhibit?

- a. patience
- c. resourcefulness
- b. creativity
- d. Rationality

10. Knowing that his knowledge on medicinal plants is limited, Dr. Raymundo made plans to meet with experts before coming up with his hypothesis. What scientific attitude does he show?

- a. resourcefulness
- c. creativity
- b. patience
- d. none of the above

Understanding

Answer the following questions:
everyday life problems?

1. Why is it important to follow the scientific method in solving both scientific and
2. How would you know which tool to use in a science experiment?
knowledge in order to be efficient in their field?
3. Why do scientists need to have the right attitudes to back up their scientific.

Product

Do the following tasks:

1. As a group of four (4) members, think of an environmental problem that you have in your school or community. Apply the steps of the scientific method in finding a solution to the problem that you identified. Make a flowchart to show your actions for each step.
2. Research on one local and one foreign scientist. Take note of the important aspects of his/her life and his/her scientific contributions. In addition, identify three (3) scientific attitudes that you think he/she possesses. Present these information through a creatively done brochure.
3. Work with a partner. Draw one laboratory tool on a sheet of bond paper. Color your work and then paste it on a used folder. Cut out your drawing into 10 pieces of different shapes. Have fun solving your puzzle and then exchange it with those of the other pairs.

